Aarav Vidhawan

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EDUCATION

University Of California San Diego — **San Diego, CA**
Bachelor of Science, Computer Engineering **| GPA: 3.85 | Provost Honors** — **2022-2026**
Relevant Coursework: Computer Architecture**,** Digital Systems, Advanced Digital Design Project, Digital Circuits, Linear Algebra, Intro to Machine Learning, Computer Organization, Data Structures & Algorithms, Operating Systems.

University Of California San Diego — **San Diego, CA**
Master of Science, Computer Engineering — **2026 - Present**

La Martiniere For Boys — **Kolkata, India**
- **Completed Grade 12, June 2022 ; ISC Result: 97.25%** — **2012-2022**
- **Completed Grade 10, June 2020; ICSE Result: 94.80%**

SKILLS

Hardware Design & Verification: SystemVerilog, Verilog, Quartus, Modelsim, Vivado, PSpice/LTSpice, Virtuoso, Vitis HLS
Languages & Frameworks: C, C++, Arduino, FreeRTOS, ESP32, STM32, SPI, I2C, USART, Python, Java, JavaScript, HTML.
Systems Programming: Threads, Synchronization, Multiprogramming, Virtual Memory
Tools: GitHub, VSCode, KiCAD, JIRA/Confluence, Lab Gear (Multimeter, Oscilloscope, Soldering Iron)

EXPERIENCE

CSE Department Tutor | Problem Solving, Communication, ARM, C — **San Diego, CA**
CSE 141/L & 140 Tutor - Computer Architecture & Digital Design — **June, 2025 – Present**
- Led weekly sections and office hours for **~50** undergrads. Prepared targeted problem sets on **ARMv8 and MIPS** pipelines, hazards, caches, memory models, and Boolean Logic, combinational & sequential logic, FSMs and RTL abstraction.
- Responded to **~40** forum questions per quarter, with an average response time under 2 hours. Co-designed and proctored midterms and final. Graded **~80** submissions using a structured rubric to ensure consistency and conceptual clarity.
- 1 on 1 discussion in Office Hours led to an average of **~15** point increase on Midterm 2 and **~10%** on the Final.
- Guided **100+ students** in **CSE 141L** through designing **custom ISAs** and **single-cycle CPUs** in SystemVerilog. Debugged datapath and control logic, improving synthesis success and comprehension of microarchitecture.

UCSD Jacobs School of Engineering | SystemVerilog, Python, Yosys — **San Diego, CA**
Undergraduate Research Assistant - SRIP - Advised by Prof. Bill Lin — **April, 2026 – Present**
- Built a compiler from trained Hopfield weights to synthesizable SystemVerilog (Espresso minimization, **graph-colored delay scheduling**); **verified at N=256, 240/240 vectors recalled**.
- Cut per-neuron logic **627–2,918 → 31–54** product terms via operating-region don't-cares, making fan-in 32 synthesizable where full enumeration is impossible.
- Showed clockless settling requires distinct delay values (equal: 0% convergence) and inertial rather than transport delay elements (100% vs 1–16%).
- **Raised storage capacity from α=0.138 to 0.5 (≥95% recall)** via margin retraining.

UCSD ITS - Information Technology Services | JavaScript, REACT, HTML, CSS, ServiceNow — **San Diego, CA**
Document And Case Management - Student Developer — **October, 2023 - June, 2024**
- Shipped **10+** production forms/workflows (Messaging Colab, Laptop Borrow Request, etc) used by **~3,000+** students and faculty, automation improvements cut manual handling from ~**30** min to ~**15** min per request(**-50%**).
- Built **3+** Analysis dashboards using ServiceNow's experience builder. Required **25+** hours of coursework.
- Built QA scripts, and performed over **100+** E2E tests of other developer's work.

PROJECTS

MacroPad - Project Lead | C++, KiCAD, LTSpice, Team Management
- Designed a modular ESP32 MacroPad with **9**, **64x64 RGB OLEDs**, mechanical keys, and **2** rotary encoders.
- Seamless desktop activation via custom, cross-platform Qt companion application. Written in C++ and Objective C.
- Implemented using serial protocol supporting profile switch < **50 ms**, and activation latency <**40 ms.**
- Designed driver circuit for the **SSD1357z** controller, integrated into custom PCB for the keyboard.
- Led a **9-person** team, sprint cadence, code review and integration.

MIPS CPU with Advanced Optimizations | SystemVerilog, Python, C
- Extended a baseline MIPS CPU with **four optimizations** inspired by research papers, and modern processors.
- Implemented **McFarling-style hybrid branch prediction** (bimodal + gshare) with BTB integration in IF.
- **MIPS R10k-style out-of-order execution engine** with register renaming, dynamic scheduling, and in-order commit.
- Integrated **hardware data prefetching** using a 4-way stream buffer for D-cache and a 1-way stream buffer for I-cache to hide miss latency. Also added **victim caches** for both, to reduce conflict misses and improve effective cache hit rate.

Sora v1.0 - Self Designed ISA and CPU | SystemVerilog, Python, Quartus, ModelSim
- Authored a 9-bit ISA (16 instr, 2 addressing modes, load-store) with an 8-bit datapath. Wrote assembler in Python.
- Achieved Hamming Distance Calculation and **double precision** multiplication using only **9 bits**. Fastest execute in the class.
- Single-cycle, with RTL in SystemVerilog, simulated in ModelSim, and resource mapped on Cyclone-V using QuartusPrime.

Nachos Operating System | Java, Nachos, MIPS, Virtual Memory

- Implemented preemptive threads with Alarm, join, interrupt-based condition variables, sleepFor and tagged Rendezvous.
- Added full multiprogramming with safe rVM/wVM, 16-FD file tables, and concurrent user-program execution.
- Built exec/join/exit with unique PIDs, parent–child synchronization, and robust process cleanup on abnormal termination.
- Designed demand-paged VM with lazy loading, swap, clock eviction, dirty-bit optimization, and page pinning.

Viterbi Encoder/Decoder | SystemVerilog, ModelSim, Quartus

- Implemented a convolutional encoder (**rate 1/2, K=7**) and a pipelined Viterbi decoder (**ACS + traceback depth 35**). Passed the end-to-end tx/rx testbench and verified **min branch metric = 0** in a no-error channel.
- Built deterministic + random error-injection. Evaluated uncorrected errors across periodic/burst patterns up to a **1/16** rate.
- Synthesized on Cyclone V. Exported RTL-viewer/netlist and wrote state-transition/branch-metric notes for reproducibility.

Nixie Tube Clock | PSpice, LTSpice, Circuit Design, CH32V

- Designed a "Nixie Clock" using 1900s vacuum tubes. used **IN-14** tubes with **11-pin** cathode/anode interface.
- Built using VSDSquadron Mini, C++ on **CH32V** platform. Built in Alarm as well, standard alarm clock layout for setting.
- Added Time Zone changing functionality, 4 buttons for 4 pre-set Time Zones.

Ultrasonic Smart Clock | PSpice, LTSpice, Circuit Design, Arduino

- Designed a "Smart Clock" that used an ultrasonic sensor to detect if the user was still sleeping or not, and decided accordingly whether or not it should ring the alarm. Used a Metro M0 Express and CircuitPython to program it.
- Added "Stop" and "Snooze" functionality. Hope to add "Time Setting" functionality in the future.
- Used Onshape to construct 3D models of the Prototype, and EagleCAD to make the circuit diagrams for the clock.

LEADERSHIP/ EXTRACURRICULAR

HKN (Eta Kappa Nu) - CSE Department Chair (Events)

- Designed and hosted **10+** technical workshops, such as the Macro Key and I2C Workshop, and also ECE Depths Seminar.
- Communicated with **15+** professors and alumni for events, technical guidance, and project support.
- Participated in weekly meetings to facilitate the creation and execution of HKN events, ranging from professional to social.

Early Learning and Cognition Lab (UCSD) - Volunteer Engineer

- Designed a child-proof "blicket" machine used to study probabilistic vs deterministic learning in children under age 5.
- Built using an Arduino, 5 hall sensors, a rotary mode switch and an RGB LED. 3D designed enclosure as well.

HOBBIES

Vinyls, Gaming, Pool, Golf, Football(Soccer), Badminton, Coffee, Food, Music, Cars, Travel